

Bridging Performance and Interpretability

Explainable Disentangled Multimodal AI for Cancer Survival Prediction

WHY DISENTANGLE?

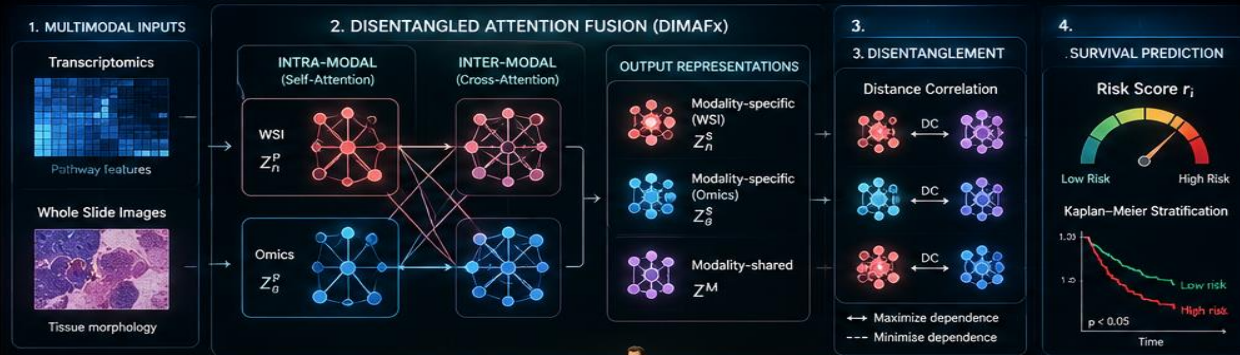


Improved interpretability
- Preserves complementary information
Better generalization

MULTIMODAL BIOLOGY



Molecular Signals — Tissue Context



ATTENTION VISUALIZATION



Cross-modal interactions highlight key relationships

TRANSPARENCY MATTERS

- Trust
- Insights
- Better Decisions

Explaining Multimodal AI – from (omics) attribution to imaging and multimodal fusion

Advanced Machine Learning and AI for Life Sciences

Wilson Silva



AI Technology for Life



- **PI of the Trustworthy AI for Life Sciences Lab**
- **Assistant Professor** – AI Technology for Life group

Previously:

- Postdoc at the Netherlands Cancer Institute
- PhD in Machine Learning for Medical Image Analysis
- BSc + MSc in Electrical and Computer Engineering

Trustworthy AI for Life Sciences

Machine learning for complex, heterogeneous and sensitive biological data. Three interconnected themes:

Explainability: What drives the model's predictions?

Privacy: How can institutions learn together without sharing sensitive data?

Generalization: will the model remain reliable in a new hospital or population?



We want AI that scientists, clinicians, and biologists can interrogate, and rely on!



Today's Roadmap

1

Why attribution
is not enough

2

Explaining
through examples

3

Interpretable
image
representations

4

Language as
explanation

5

Interpretable
Multimodal
Fusion

Main benefit of DL over classical ML

- Why did deep learning replace classical machine learning in so many domains?



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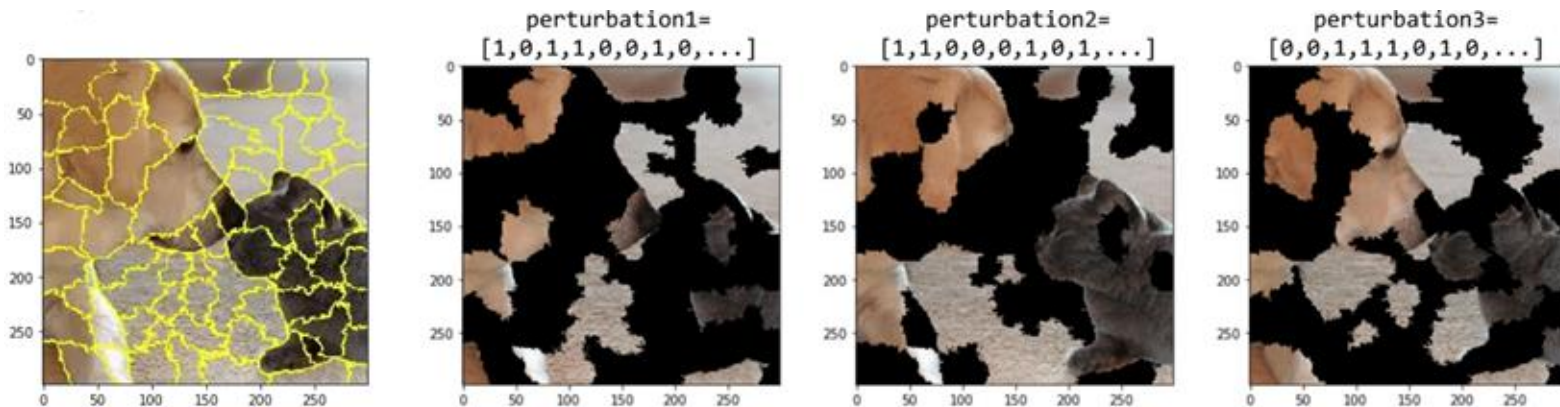
Event code

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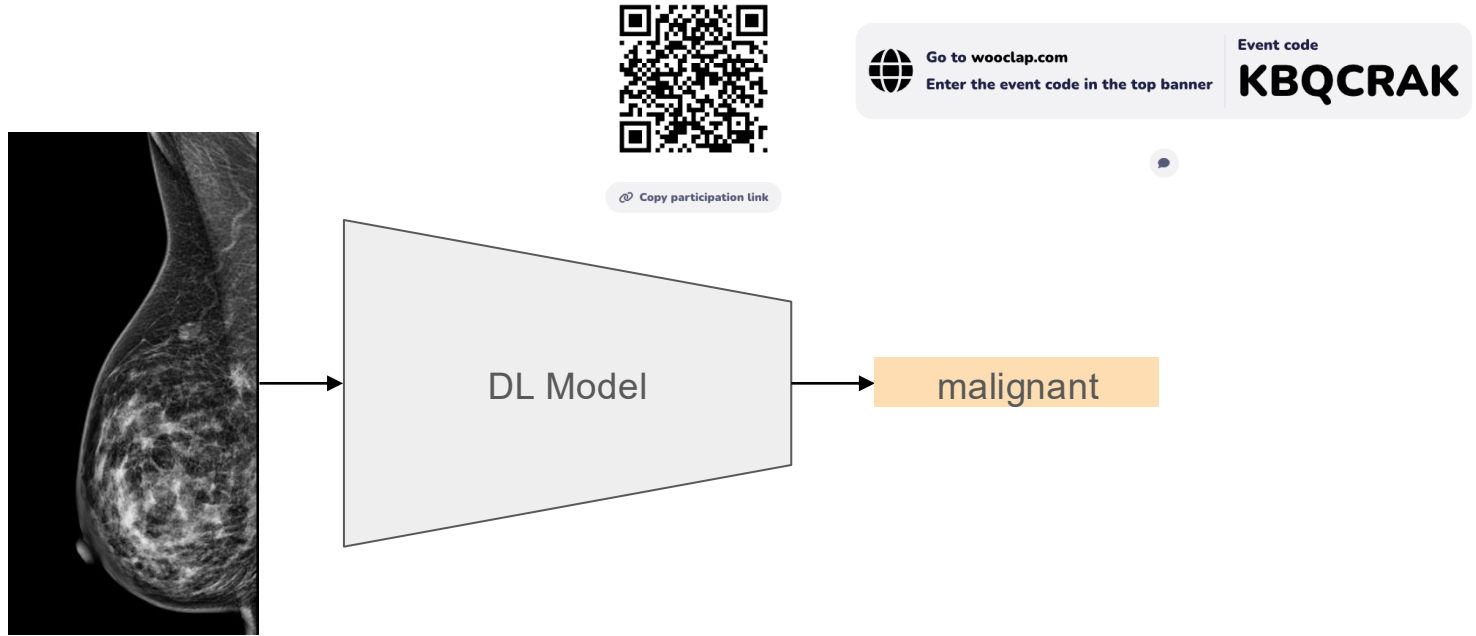
Towards Interpretable Multimodal AI

- Feature attribution explains *which* features mattered.
- But what if the features themselves are not meaningful?
- How do we design models whose internal representations are interpretable?



A model predicts cancer

- How would you explain this prediction to a radiologist?



Feature attribution is useful, but incomplete

Feature attribution tells us:

- Which features influenced the prediction
- Whether a feature increases/decreases the output
- Local importance around a prediction

⚠ But it does not tell us:

- What visual concept was recognized?
- Does the reasoning resemble known cases?
- Are the learned representations meaningful?
- How do representations from different modalities interact?

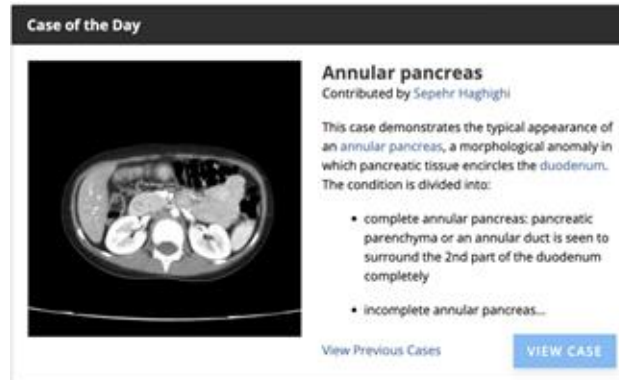
Connecting to real evidence

Humans (often) reason by comparison:

"Have I seen something like this before?"

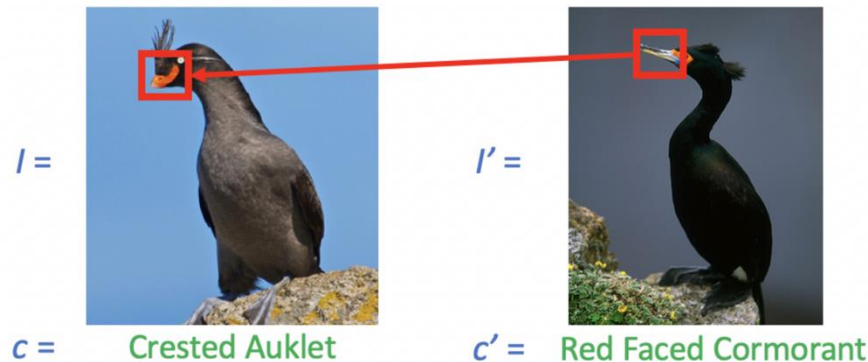
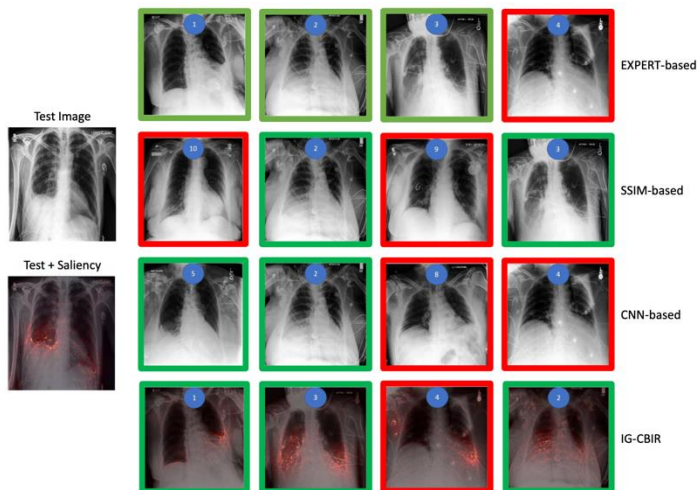
→ **Example-based explanations support this by retrieving similar cases.**

- Aligns with medical practice – radiologists look for similar cases in databases (e.g., Radiopaedia)



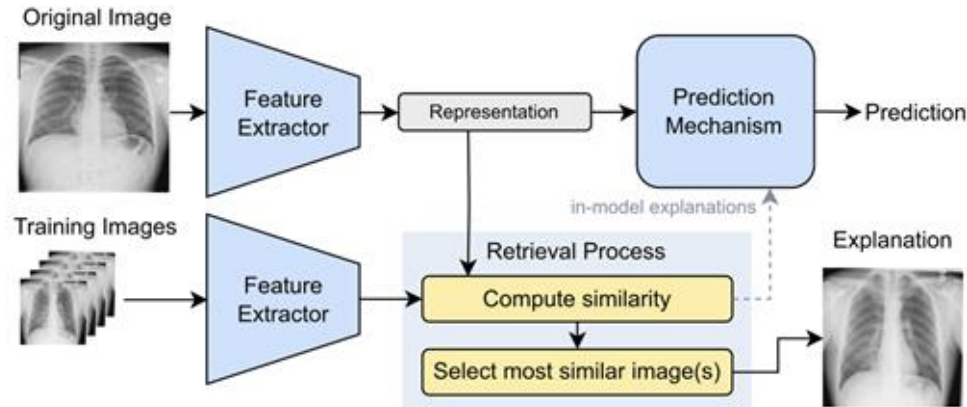
Example-based Explanations

- Types of **example-based** explanations:
 - based on **real data** - derived from actual patient cases
 - **counterfactual** explanations - use synthetic data to show *what-if* scenarios



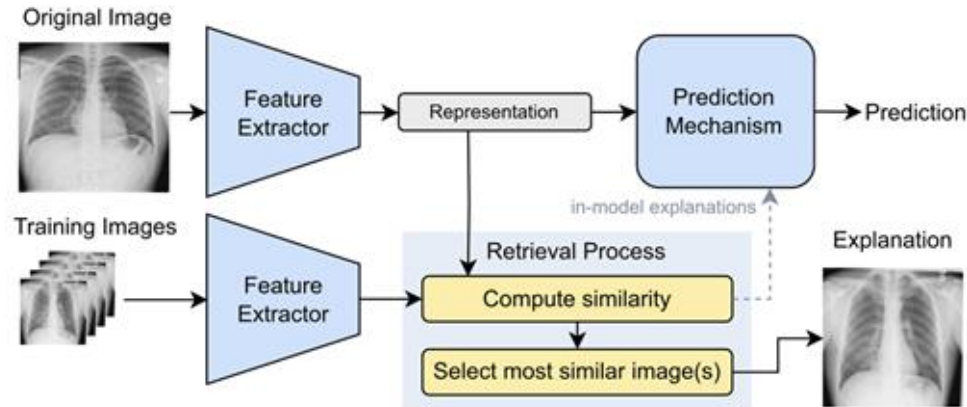
Explanations based on Real Data

- Explanations based on Real Data:
 - via **intrinsically interpretable** models
 - or in a **post hoc** manner



Explanations based on Real Data

- **In-model approaches** (directly **use training images** to make a **decision** about **test image**):
 - **prototype-based** (classify images based on **similarity** to learned **prototypes**)
 - **distance-based** (**map** images to a **semantic latent space** and classify based on **distance**)



How should we measure similarity?



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(a) Test Image



Retrieved Image 1



Retrieved Image 2

How should we measure similarity?



(a) Test Image



(b) Least Similar Image



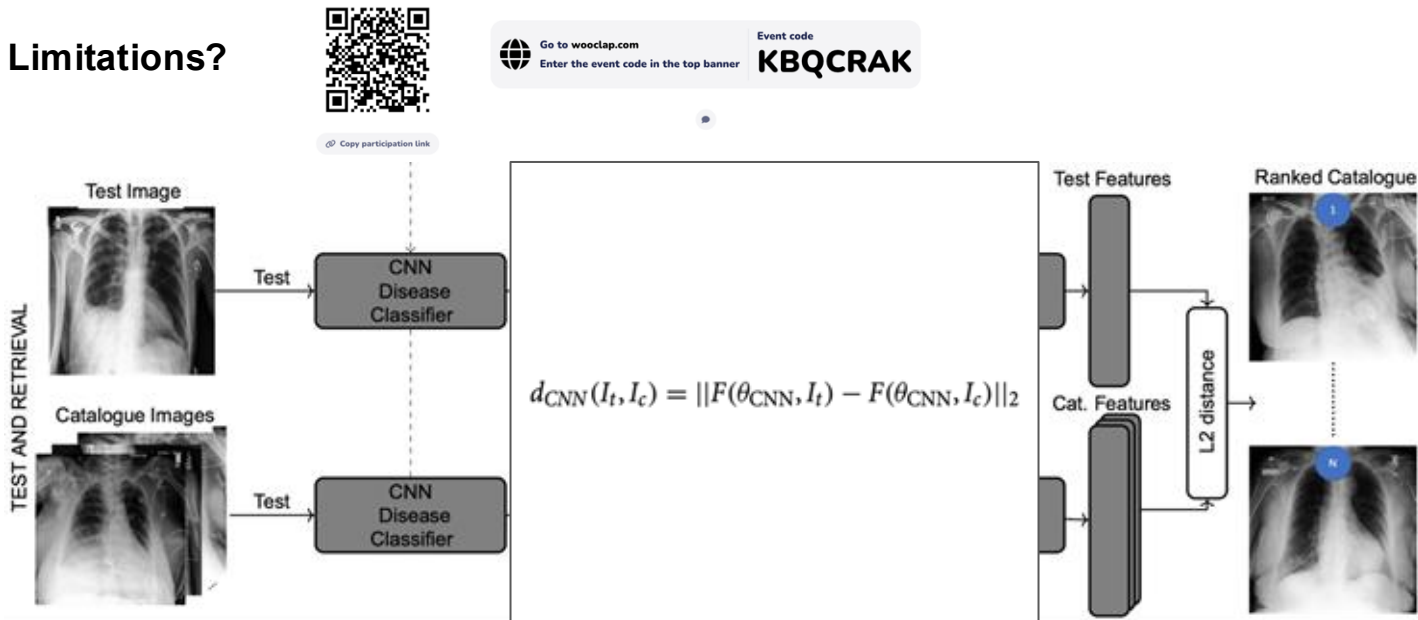
(c) Most Similar Image

What about relying on a classifier?

- **CNN disease classifier**

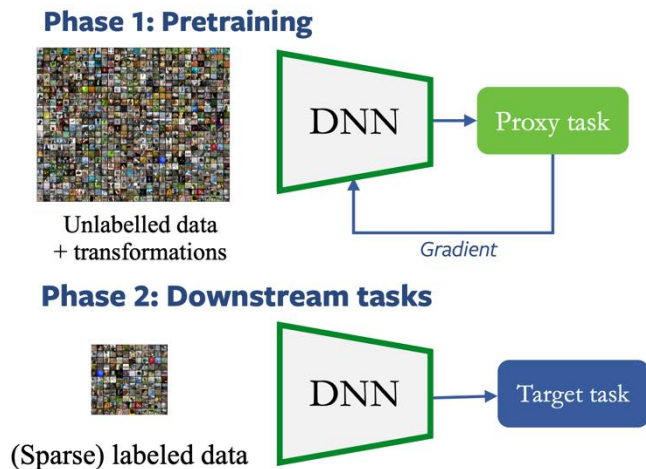
- **CNN-based disease classifier** extracts **semantic** representations
- **Shown to better approximate radiologists' similarity judgements**

- **Limitations?**



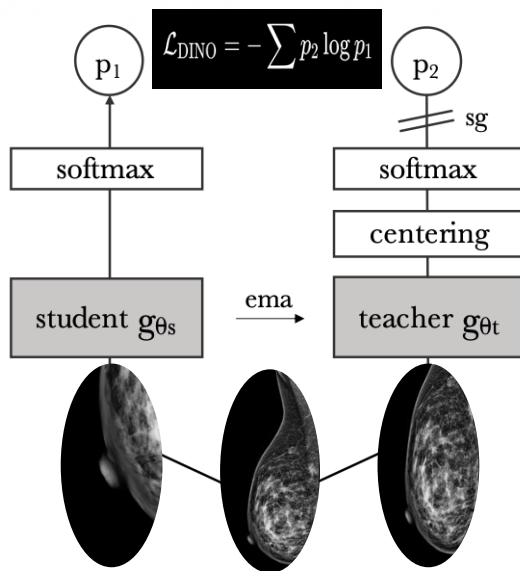
Learning representations without labels

- Learning useful representations **traditionally** requires **labelled data**
- **Self-supervised learning** replaces manual labels with proxy learning objectives
- Pretrained **representations** can then be **transferred** to **downstream tasks**



self-Distillation with NO labels (DINO)

- Different **augmentations** of the same **image** should produce **similar representations**
- Teacher receives NO gradient updates; only student learns directly



UNI – Self-supervised Model for Pathology

$$\mathcal{L} = \mathcal{L}_{\text{DINO}} + \mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{reg}}$$

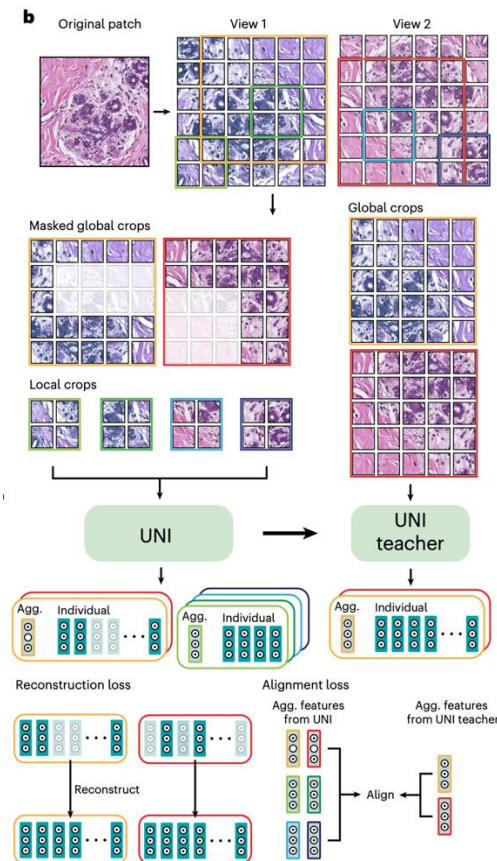
- DINO – self-distillation loss (=alignment loss)
 - minimizes the predictive categorical distribution (global repres.)

$$\mathcal{L}_{\text{DINO}} = - \sum p_{\text{UNI teacher}} \log p_{\text{UNI}}$$

- DINOv2 – adds masked image modelling loss (=reconstruction loss)
 - minimizes the predictive categorical distribution for masked tokens

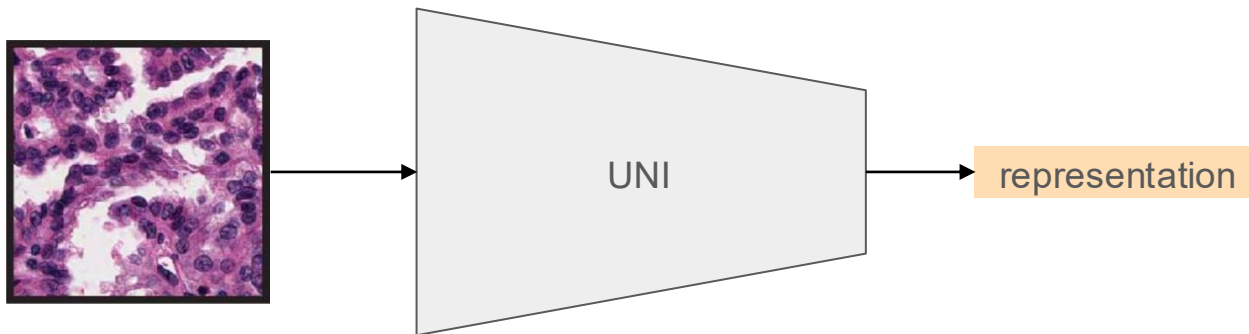
$$\mathcal{L}_{\text{recon}} = - \sum_{m \in \mathcal{M}} p_{\text{UNI teacher}}^{(m)} \log p_{\text{UNI}}^{(m)}$$

- UNI – applies DINOv2-style pretraining to WSI data
 - Learns general-purpose local tissue representations



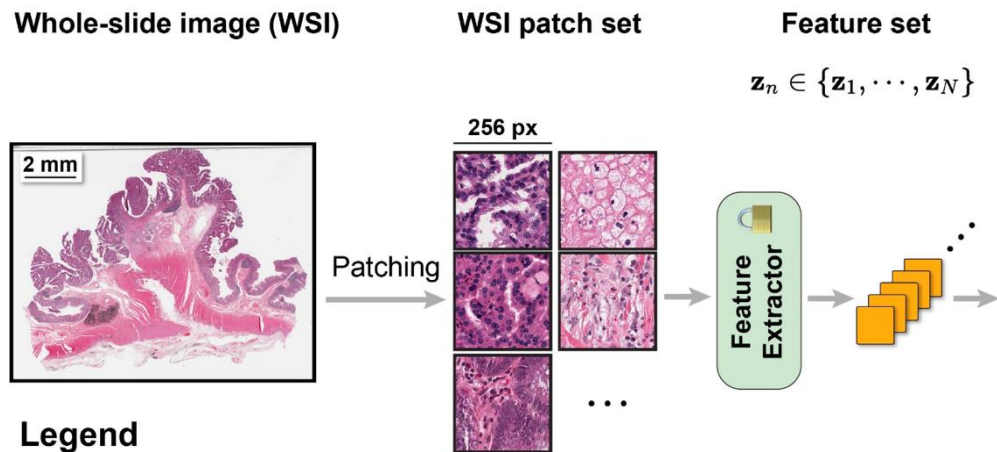
From task-specific to general representations

- **Task-specific** models learn **features** optimized for **one prediction task**
- **Self-supervised** models capture **broader patterns** of tissue variation
- The same **pretrained representation** can support **multiple downstream tasks**

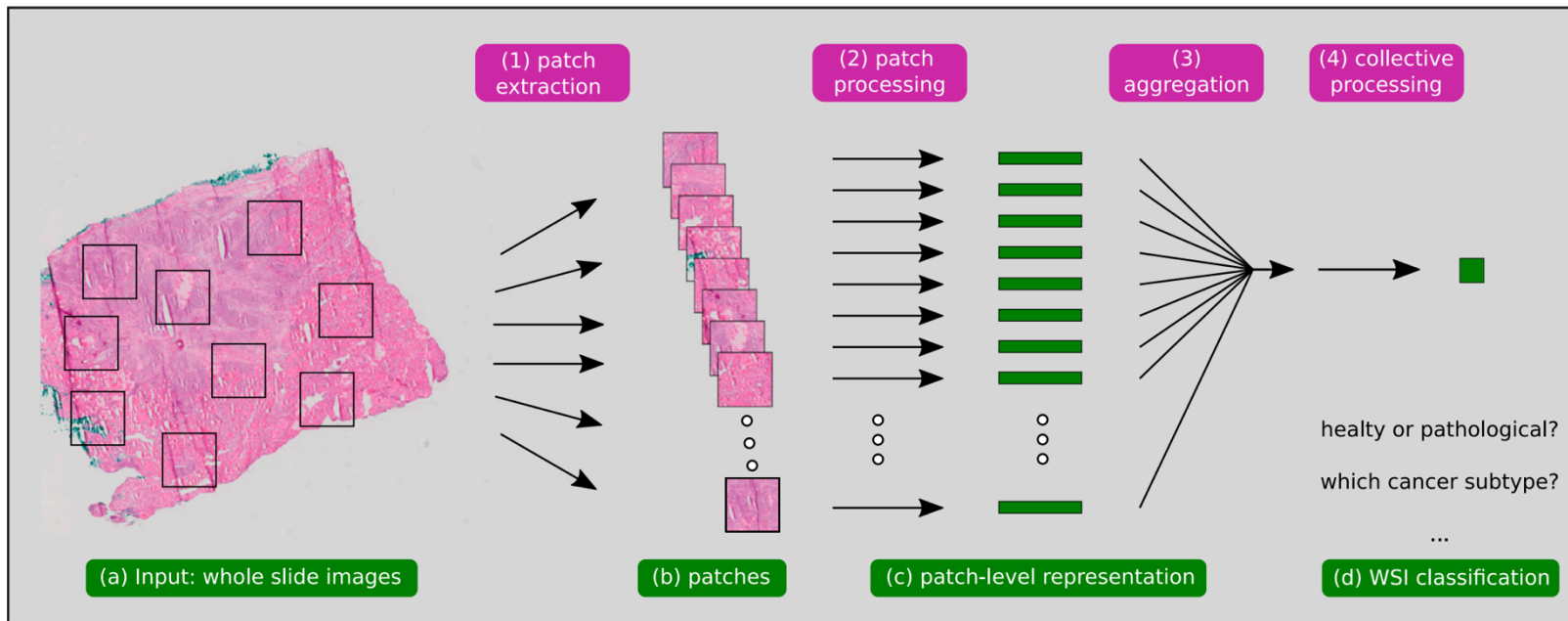


From local representations to WSI understanding

- **UNI** produces **local tissue representations**
- **Diagnosis** depends on **global tissue** organization
- Thousands of **local representations** must be **combined**



From local features to slide-level prediction



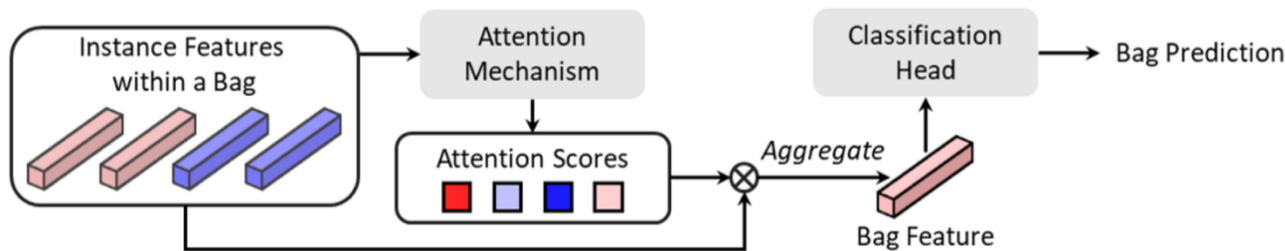
Multiple Instance Learning for Whole Slide Images

- WSI are too large to process end-to-end
 - split the slide into many patches (instances)
 - one collection forms a bag (labelled)
 - MIL learns to identify informative patches
 - **Attention** assigns an **importance score** to each patch

$$x_{\text{bag}} = \sum_{i=1}^N a_i x_i$$

$$\hat{y} = \text{softmax}(W x_{\text{bag}} + b)$$

$$\mathcal{L}_{\text{ABMIL}} = - \sum_{c=1}^C y_c \log(\hat{y}_c)$$

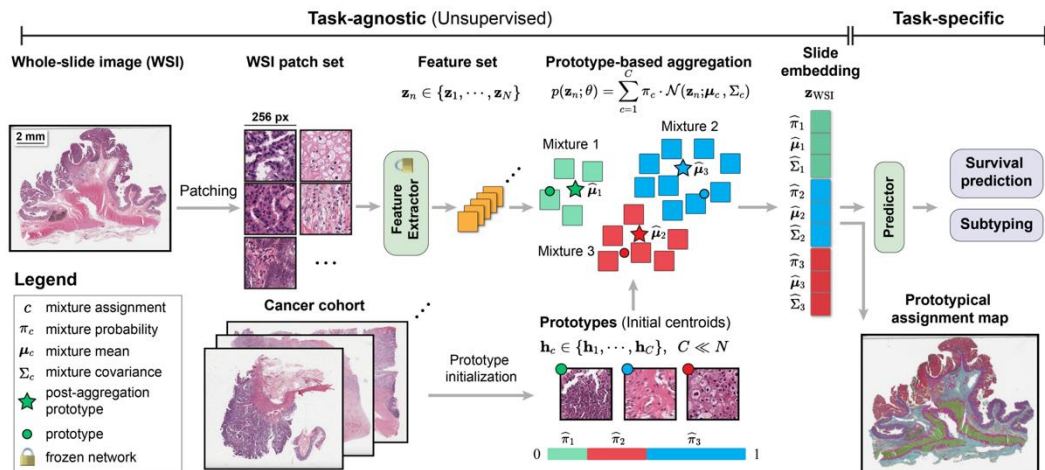


From attention to interpretable morphology

- **Attention** tells us *where*
- But **not what morphological patterns** define the slide
- We **need** a structured and **interpretable slide representation**

PANTHER – Prototype Aggregation

- **Scales prototype** reasoning to **whole slides** – thousands of patches are summarized using a much smaller number of mixture components
- **Unsupervised slide representation** – prototypes are learned without downstream task labels
- **Richer** than “this looks like that” – also tells you how much of each pattern is present and how variable it is across the slide



PANTHER – How the representation is created

STEP 1

Each patch is softly assigned to a set of shared morphological prototypes (modelled as a Gaussian mixture)

1

Initialize Shared Prototypes

- cluster all patch embeddings
- obtain C common morphological patterns
- shared across all slides
- *a dictionary of tissue patterns*

2

Estimate the mixture for each slide

- Each patch is softly assigned to multiple prototypes
- estimate per-slide:
 π : how much of each pattern
 μ : mean feature representation
 Σ : variability

3

Build Slide Representation

- concatenate all prototype statistics
- fixed-length, interpretable embedding

PANTHER – Slide embedding and interpretability

STEP 2

Each slide is represented through per-component prevalence, feature description and variation.

Two interpretability tools

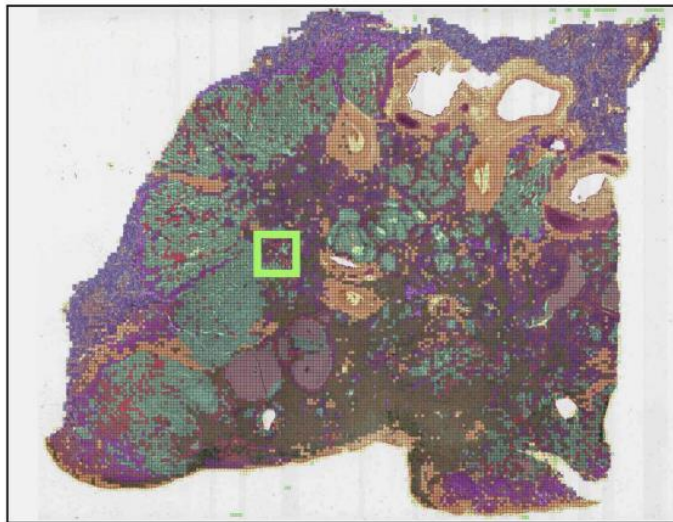
A Prototypical assignment map

Shows where each pattern appears in the slide.

B Prototype distribution and posterior heatmaps

Shows how much of each pattern is present

Prototypical Assignment Map (C=16)



Result: PANTHER turns a gigapixel image into something a human can reason about.

Break (5 minutes)



Image Captioning as Explanation

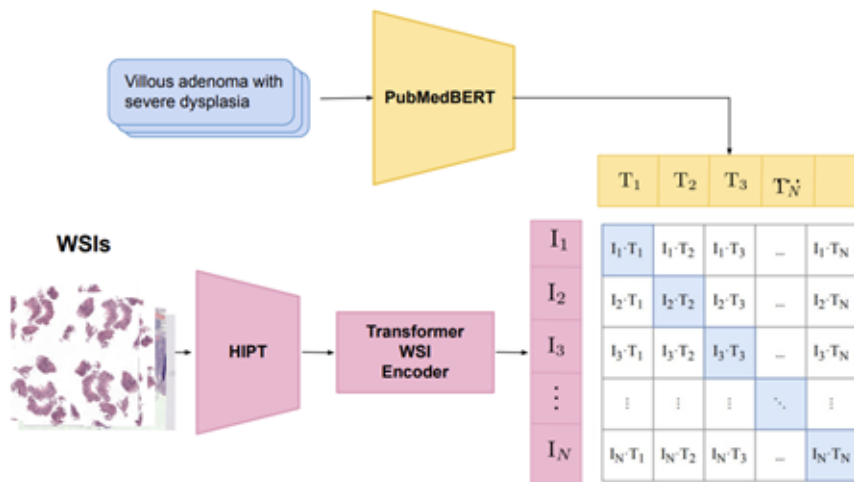
- The **explanation becomes a sentence** instead of a heatmap
 - describes visual content using **natural language**
 - combines a **visual model** with a **language model**
 - **image and text representations** can be **aligned** in **shared embedding space**

Image Captioning as Explanation

- How do we **generate a sentence**?
 - **input**: image features
 - **output**: a sequence of words
 - **challenge**: sequentially predicts words while conditioning them on visual information

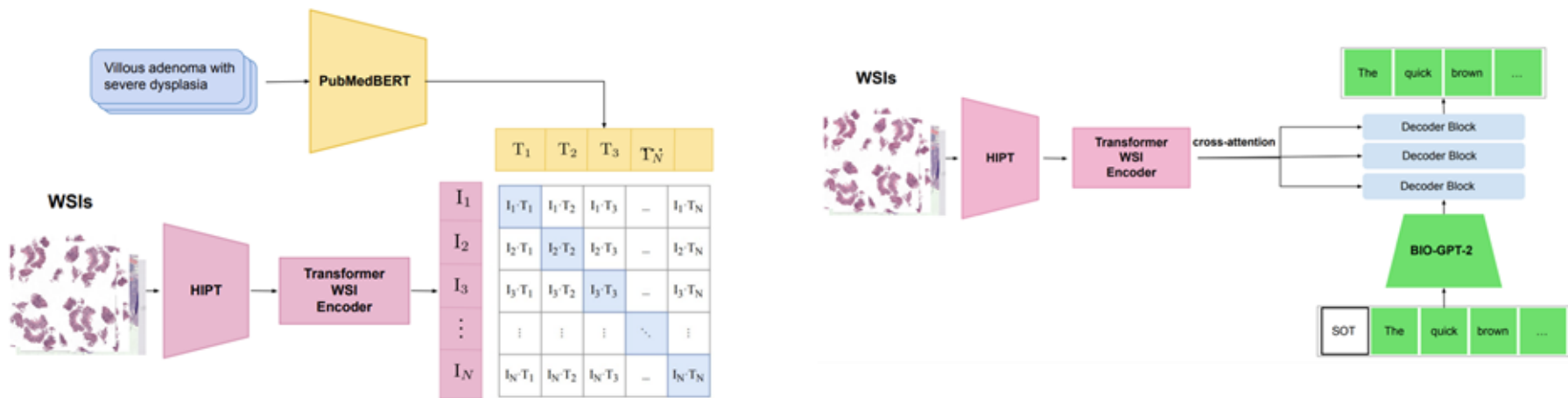
Whole-slide image captioning

- Knowledge from **pretrained vision transformers** can be adapted to downstream tasks
- Weakly-supervised **transformer-based model** for **classification** and pathology **report generation**



Whole-slide image captioning

- **HIPT encodes WSI** and **PubMedBERT encodes medical text**
 - 1st stage: contrastive WSI-caption pre-training
 - 2nd stage: decoder blocks are conditioned on the WSI embeddings



Whole-slide image captioning

- training with **GPT-3.5-cleaned captions** improved diagnostic accuracy and **caption quality**
- captioning model achieves accuracy close to a supervised classifier
- prompt-style caption preprocessing normalized structural differences across datasets

Unpretrained caption model	Pre-trained caption model	GPT-3.5 cleaned pre-trained caption model	WSI supervised classifier
0.65 (± 0.20)	0.70 (± 0.21)	0.73 (± 0.15)	0.76 (± 0.16)

Original Caption	GPT-3.5 Cleaned	Generated Caption
biopsies distal colon: chronic inflammation, in partially active and slightly histiocytary. no specific characteristics. the microscopic preparations from elsewhere have been requested for revision.	chronic inflammation, no specific characteristics.	no abnormalities, no dysplasia or malignancy. cyclic inflammation.
biopt colon transversum: adenocarcinoma.	adenocarcinoma.	Metastasis of adenocarcinoma best suited to primary process.
1) fragments of tubular adenoma with high degree dysplasia.	tubular adenoma with high degree dysplasia.	adenocarcinoma on villous adenoma. no lymphovascular invasion is identified. enced enced enced ED ED ED ED

Image captioning

- Potential **limitations** of image captioning



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Potential limitations of image captioning

- **may not** accurately **reflect** the model's **decision-making process**
- generated text might **align** with **human expectations rather** than model **reasoning**
- may produce **descriptive rather than explanatory** outputs
- may **introduce biases** that distort explanations

B E Y O N D

PANTHER

PANTHER summarises a WSI into a prototype distribution.

What if we also had the patient's molecular profile?

PANTHER gives us...

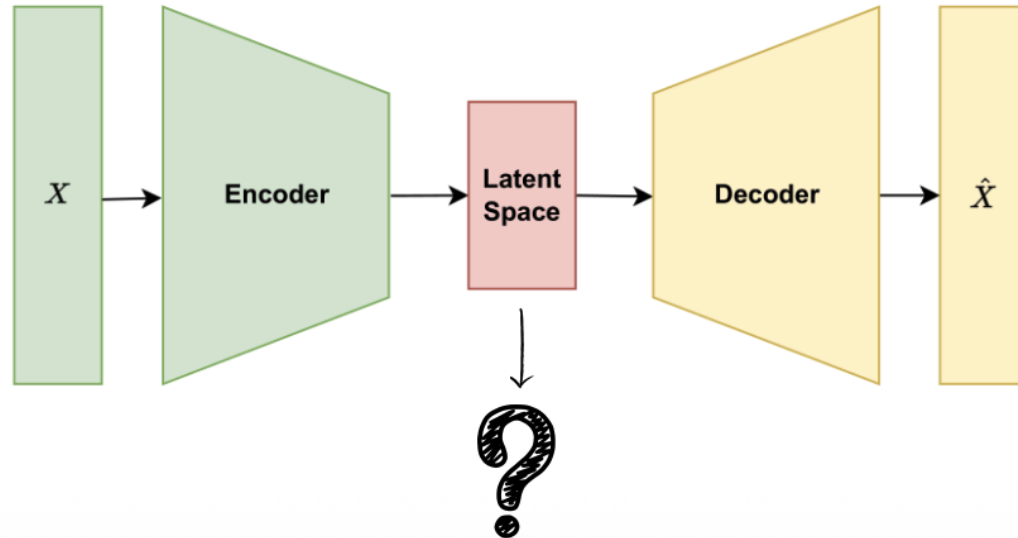
- ✓ WSI represented as a mixture of morphological prototypes
 - ✓ Per-prototype proportion (π), feature description (μ) and spread (Σ)
 - ✓ Spatial assignment maps connect prototypes back to tissue
-

But histology alone has limits...

- ! Molecular processes are only partially observable in morphology
- ! Transcriptomics provides complementary biological information

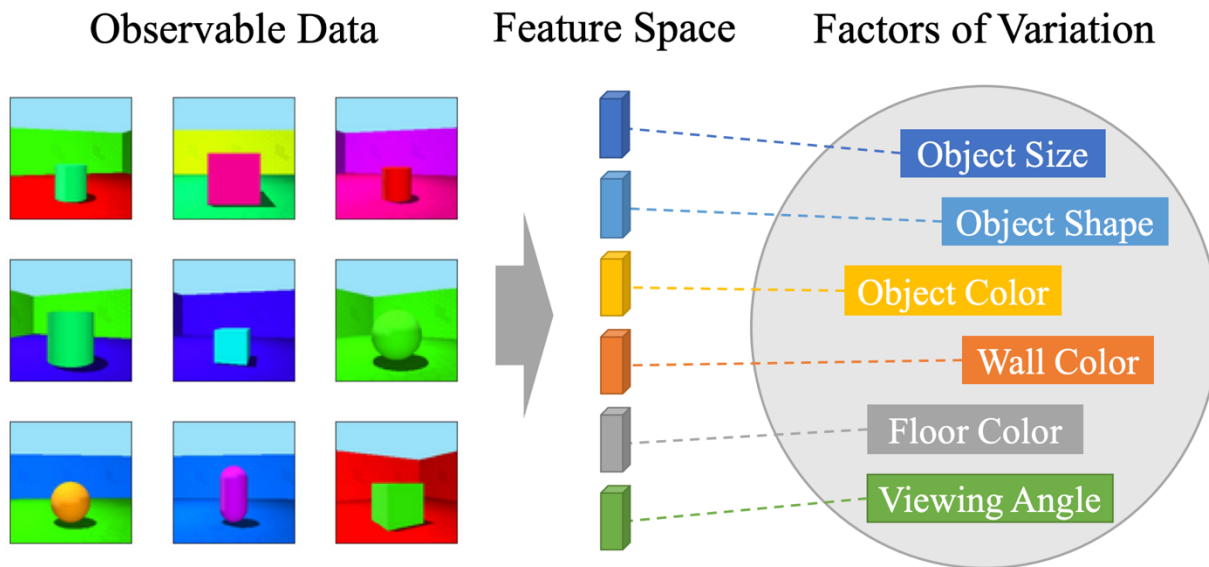
Representation Learning

- Map high-dimensional observations into a **latent representation**
- **Retain information useful** for downstream tasks
- Individual latent dimensions do **not necessarily have** an **interpretable meaning**



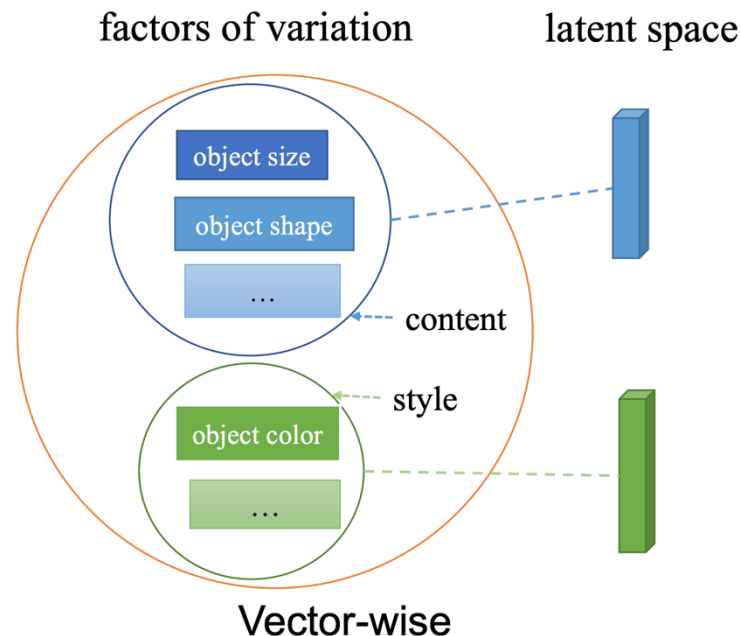
Disentangled Representation Learning (DRL)

- Encourage **different latent components** to capture **distinct factors of variation**
- Can support more interpretable and robust representations



Shared and modality-specific representations

- **Content-Style disentanglement** networks (CSD)
 - Content: spatial information
 - Style: appearance information
- **Often** used for **multimodal data**
 - modality-shared
 - modality-specific



Explaining Multimodal Data

How would you explain a model that makes decisions based on multiple modalities

Where you need to know what it attends to, what each part represents, and how much each part contributes?



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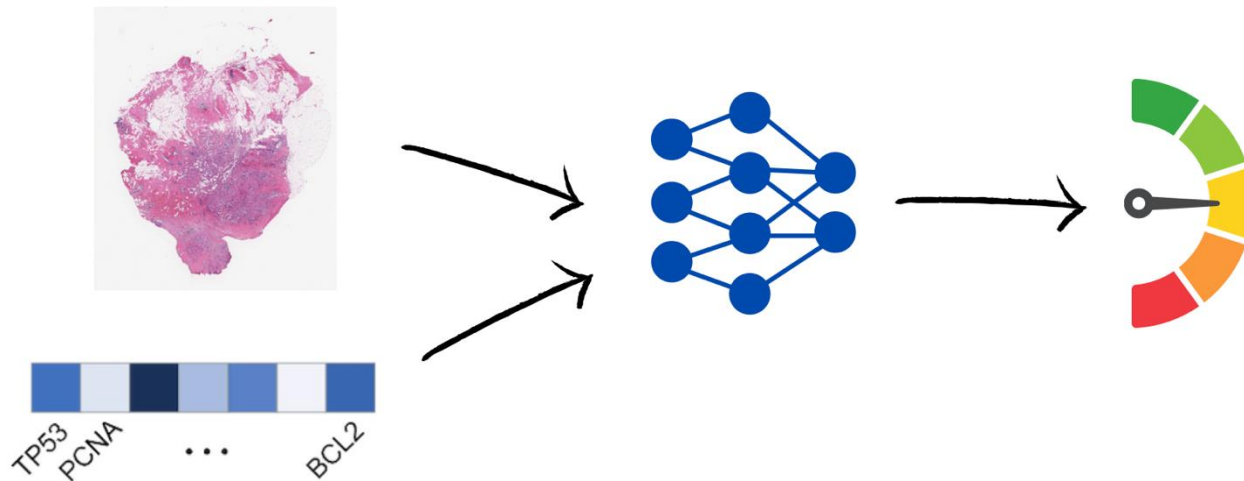
Event code

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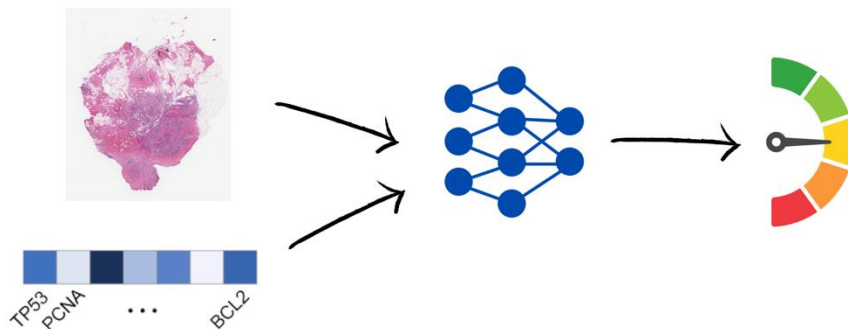
Multimodal Cancer Survival Prediction

- Predict patient risk by **combining histopathology** and **transcriptomics**
- The **modalities** encode **complementary** aspects of the tumour
- **Fusion** may **improve prediction**, but it can make the process **harder to interpret**



What each modality contains

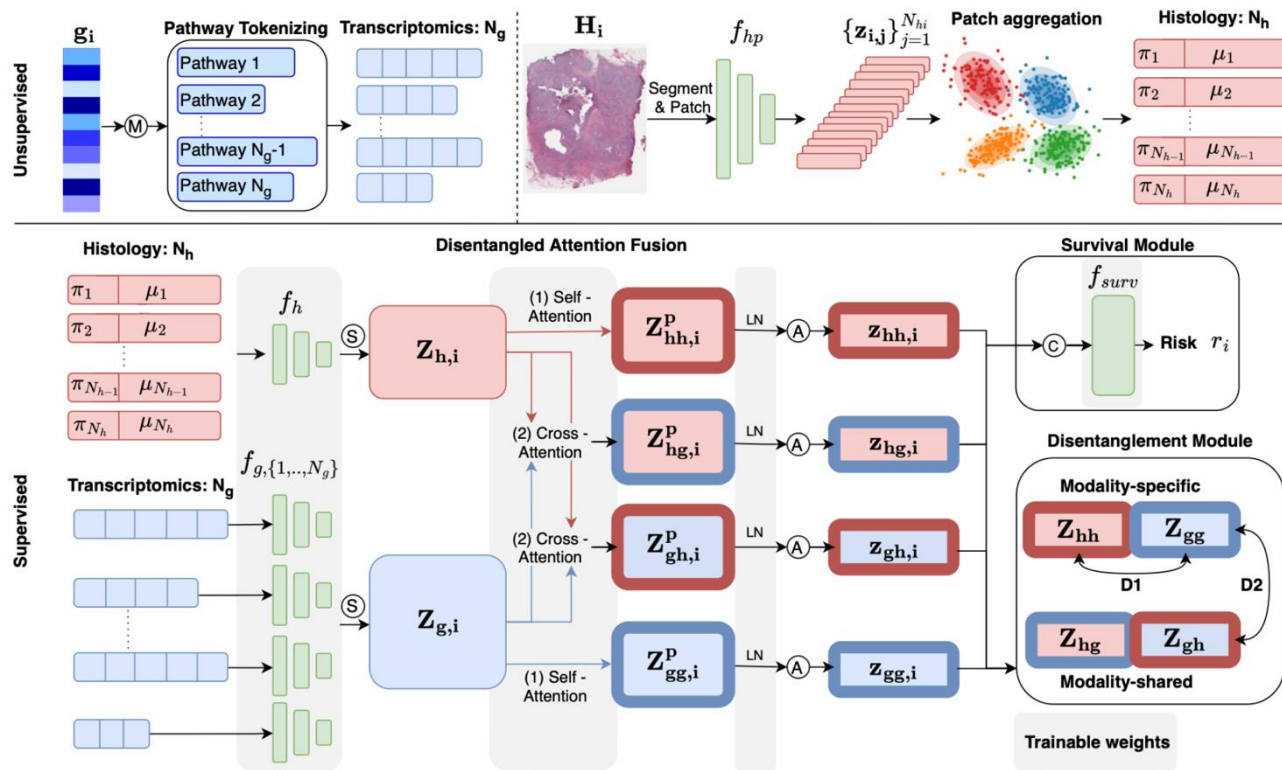
- **Whole Slide Images** (Gigapixel images)
 - Spatial organization of the tumour and surrounding tissue
 - Cellular morphology and tissue heterogeneity
- **Gene Expression** (High dimensional tabular data)
 - Global molecular activity in the tumour
 - Information about active biological processes



DIMAF roadmap

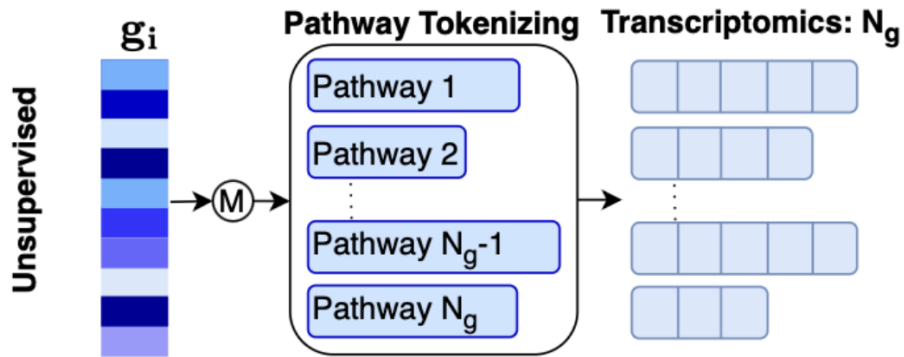
- Generate **interpretable** unimodal **representations**
- **Disentangle** the fusion between **WSI** and **Transcriptomics**
 - Modality-specific representations
 - Modality-shared representations
- Promote **disentanglement** with Distance Correlation (DC)
- **Assess** relative **contribution** to survival prediction by using **SHAP**

DIMAF architecture



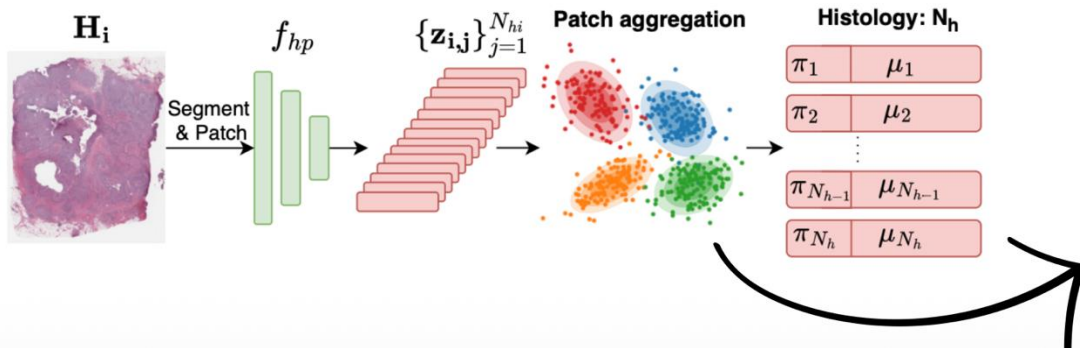
Transcriptomic feature extraction

- Biological **pathways** define **groups of genes** involved in **related processes**
- Gene-expression data are organized according to these pathways
- Each pathway is represented as a transcriptomic token



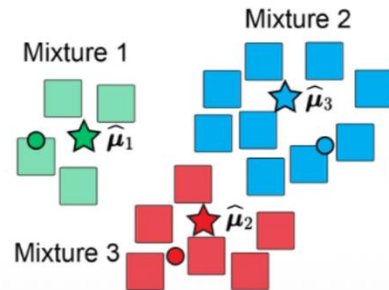
Histopathology feature extraction

- **Patch embeddings** are **softly assigned** to **morphological components**
- **Prevalence**: estimated proportion of patches associated with component
- **Mean**: morphological prototype of that component



Prototype-based aggregation

$$p(\mathbf{z}_n; \theta) = \sum_{c=1}^C \pi_c \cdot \mathcal{N}(\mathbf{z}_n; \boldsymbol{\mu}_c, \Sigma_c)$$



DIMAF: disentangled attention fusion

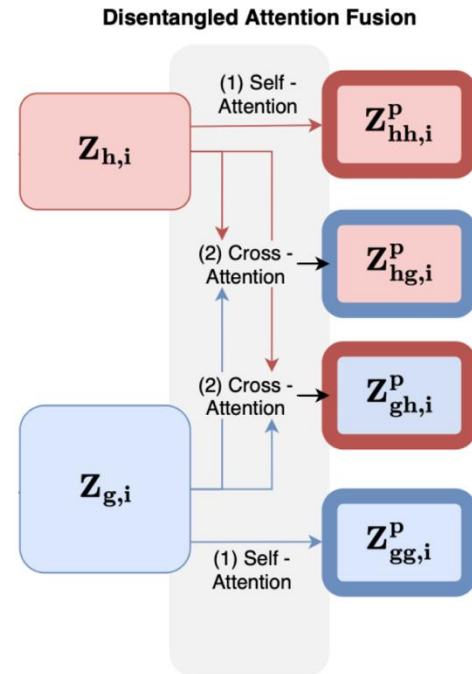
- **self-attention** to create **modality specific** representations

$$\mathbf{Z}_{\{gg, hh\}, i}^p = \sigma \left(\frac{(\mathbf{Z}_{\{g, h\}, i} \mathbf{W}_{Q1}^{\{g, h\}})(\mathbf{Z}_{\{g, h\}, i} \mathbf{W}_{K1}^{\{g, h\}})^T}{\sqrt{d}} \right) (\mathbf{Z}_{\{g, h\}, i} \mathbf{W}_{V1}^{\{g, h\}})$$

- **cross-attention** to create **modality shared** representations

$$\mathbf{Z}_{\{gh, hg\}, i}^p = \sigma \left(\frac{(\mathbf{Z}_{\{h, g\}, i} \mathbf{W}_{Q2}^{\{h, g\}})(\mathbf{Z}_{\{g, h\}, i} \mathbf{W}_{K2}^{\{g, h\}})^T}{\sqrt{d}} \right) (\mathbf{Z}_{\{g, h\}, i} \mathbf{W}_{V2}^{\{g, h\}})$$

- **model** can learn to **focus** on **different aspects** of the data, facilitating disentanglement



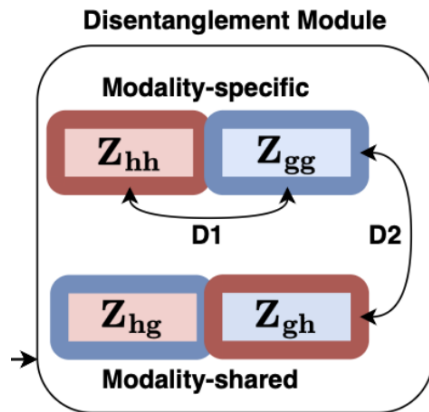
DIMAF: disentanglement module

- **distance correlation (DC)**
- measure **dependence** of two **representations**

$$\mathcal{L}_{dis} = DC(\mathbf{Z}_{gg}, \mathbf{Z}_{hh}) + DC([\mathbf{Z}_{gg}, \mathbf{Z}_{hh}], [\mathbf{Z}_{hg}, \mathbf{Z}_{gh}])$$

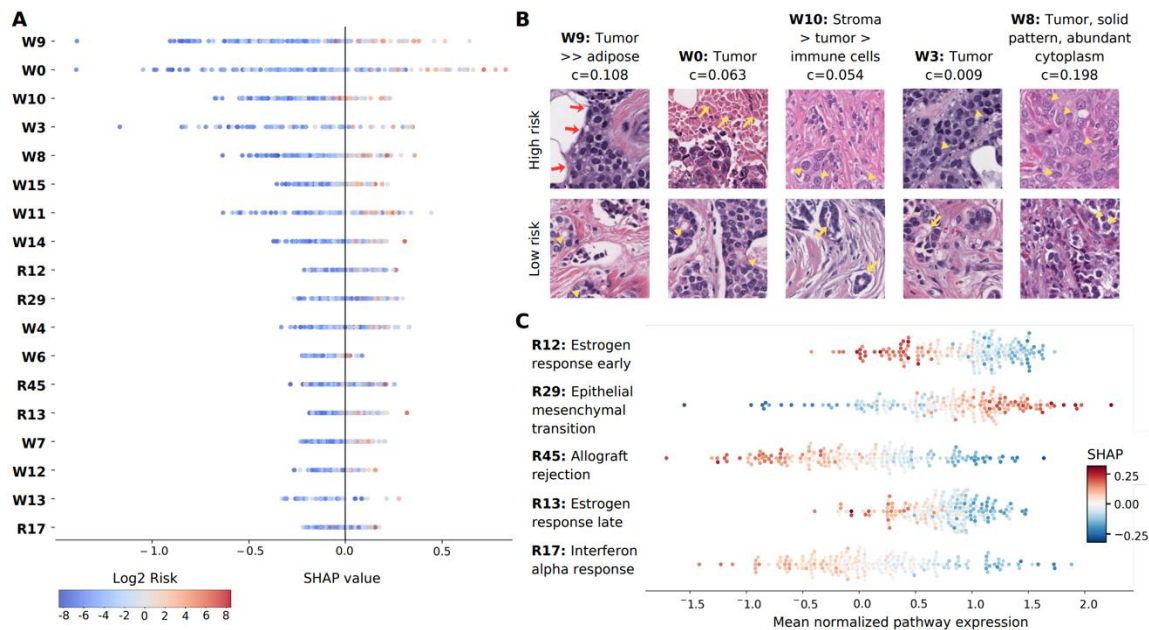
$$DC(\mathbf{Z}_1, \mathbf{Z}_2) = \frac{dCov(\mathbf{Z}_1, \mathbf{Z}_2)}{\sqrt{dCov(\mathbf{Z}_1, \mathbf{Z}_1)dCov(\mathbf{Z}_2, \mathbf{Z}_2)}}$$

- **distance covariance (dCov)** measures dependence by comparing pairwise distance



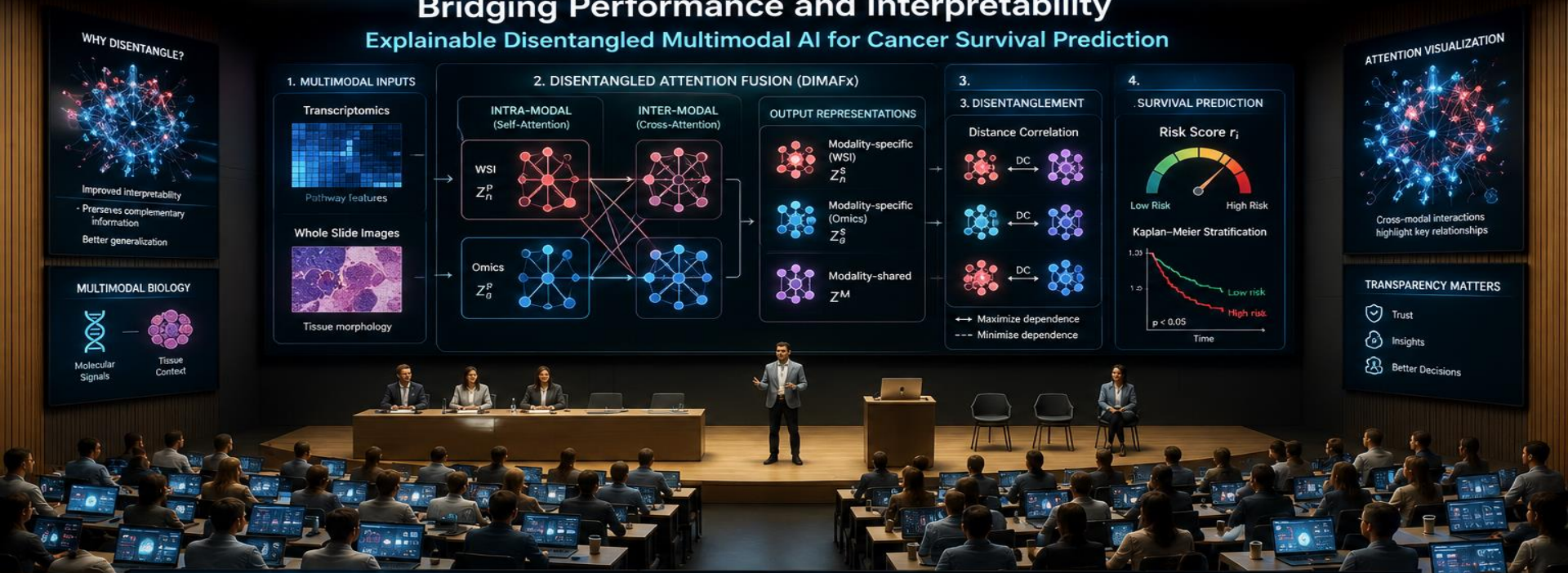
From DIMAF to DIMAFx: feature contribution

Model	Repr.	BRCA	BLCA	LUAD	KIRC	Avg.
DIMAF _{-dis}	Specific	0.513 $\pm$ 0.033	0.548 $\pm$ 0.028	0.538 $\pm$ 0.028	0.517 $\pm$ 0.024	0.529
	Shared	0.487 $\pm$ 0.033	0.452 $\pm$ 0.028	0.462 $\pm$ 0.028	0.483 $\pm$ 0.024	0.471
DIMAF	Specific	0.180 $\pm$ 0.037	0.273 $\pm$ 0.038	0.257 $\pm$ 0.027	0.302 $\pm$ 0.043	0.253
	Shared	0.820 $\pm$ 0.037	0.727 $\pm$ 0.038	0.743 $\pm$ 0.027	0.698 $\pm$ 0.043	0.747



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Explaining Multimodal AI – from (omics) attribution to imaging and multimodal fusion

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AI Technology for Life